Topic 35 — Atmosphere

Covers syllabus (click to expand)

Structure of the Atmosphere | Atmospheric Composition | Atmospheric Layers | Troposphere (क्षोभमंडल) | Stratosphere (समतापमंडल) | Mesosphere (मध्यमंडल) | Thermosphere | Exosphere

Sources baked in: NCERT Geo 11 Ch 8; MoEFCC/CPCB context; UPPCS/UPSC PYQs **Weight:** ★★★ High — ozone = stratosphere; heating by longwave; $N_2 \neq$ GHG **Last verified:** August 2026

Current Affairs (this topic)

- Atmosphere layer and composition facts are stable; WMO GHG / ozone bulletins are light CA.
- Troposphere weather vs stratosphere ozone remains the classic layer trap.

35.1 Structure of the Atmosphere

Identity: Gaseous envelope | Temp-based layers | Heated from below

- The **atmosphere** is the gaseous envelope held by Earth's gravity.
- About **99% of atmospheric mass** lies within roughly **32 km** of the surface.
- Structure is mainly classified by the **vertical temperature profile** into five layers (see §35.3–35.8).
- By composition: **homosphere** (well mixed to ~**80–90 km**) and **heterosphere** (gases separate by mass above that).
- Homosphere thermal layers are **troposphere + stratosphere + mesosphere**. The **ionosphere** is **not** included in the homosphere set in UKPCS-style stems.
- Sea-level pressure ≈ **1013 mb (1 atm)**; pressure and density fall rapidly with height.
- The Sun's **shortwave** energy mainly heats the **surface**; the surface emits **longwave infrared** that heats the lower atmosphere from below.

PYQ — UPPCS Prelims 2022, Q138

The Earth's atmosphere is mainly heated by which one of the following?

Options: A. Long-wave terrestrial radiation B. Scattered solar radiation C. Reflected solar radiation D. Short-wave solar radiation

Show answer

Ans: A

> **Logic:** “Atmosphere heated mainly by direct shortwave” is **false**.

35.2 Atmospheric Composition

Identity: N_2 78% | O_2 21% | Ar ~1% | CO_2 ~0.04% | Variable gases | Mixture | Radon trap

- Air is a **mixture** of gases, not a single compound or element.
- **Dry air near the surface: Nitrogen ~78.08%, Oxygen ~20.95%, Argon ~0.93%, CO_2 ~0.04%.**
- **Permanent gases** (N_2, O_2, Ar) stay roughly constant in the homosphere.
- **Variable gases** include water vapour (0–4%), CO_2, CH_4, O_3, and aerosols.
- Noble gases present in air include **helium**, **neon**, and **argon**. **Radon** is **not** treated as a normal constituent of atmospheric air in this composition set.
- Plants release oxygen through photosynthesis. If plants and trees vanished, atmospheric **oxygen** would decline.
- The **Amazon rainforest** is estimated to produce about **20 percent** of the oxygen generated by terrestrial photosynthesis, which is why it is often called the “lungs of the Earth.”
- **N_2 is not a greenhouse gas**; **N_2O** is. Do not confuse N_2 with nitrogen oxides or nitrous oxide.
- **Argon is not a GHG** (also covered in Topic 34 greenhouse-gas questions).
- **Water vapour** is the strongest natural tropospheric greenhouse gas by volume, but mainly a climate **feedback**.
- Dust/aerosols scatter light (blue sky / red sunset).

PYQ — UPPCS Prelims 2020, Q139

Which is NOT a greenhouse gas found naturally in the atmosphere?

Options: A. Nitrogen oxide B. Carbon dioxide C. Methane D. Ozone

Show answer

Ans: A — CO_2, CH_4, and O_3 are natural GHGs here. Nitrogen oxides are not treated as that major natural GHG set.

PYQ — UPPCS Prelims 2021, Q134

Diffusion of light in the atmosphere takes place due to

Options: A. Carbon dioxide B. Dust particles C. Helium D. Water vapours

Show answer

Ans: B

Gas	Approx. %	Key fact
N_2	78%	Not GHG

Gas	Approx. %	Key fact
O_2	21%	Respiration
Ar	0.93%	Inert; not GHG
CO_2	0.04%	GHG
H_2O	0–4%	Variable

> **Logic:** Composition % ≠ greenhouse importance (CO_2 is tiny but powerful).

35.3 Atmospheric Layers

Identity: Five temperature layers | Pauses reverse the trend

- Five layers from surface outward: **Troposphere** → **Stratosphere** → **Mesosphere** → **Thermosphere** → **Exosphere**.
- Boundaries: **tropopause, stratopause, mesopause** — where the temperature trend reverses.
- Temperature does **not** fall continuously with height through all layers.

Layer	Approx. height	Temp trend	Signature
Troposphere	0–8/18 km	↓ upward	Weather
Stratosphere	~12–50 km	↑ upward	Ozone
Mesosphere	~50–85 km	↓ upward	Meteors; coldest
Thermosphere	~85–600 km	↑ upward	Aurora / ISS
Exosphere	~600 km+	Transition	Space merge

> **Logic:** Learn **feature** ↔ **layer** matching, not only names.

35.4 Troposphere

Identity: Lowest layer | All weather | Lapse rate ~6.5°C/km

- The **troposphere** is the lowest layer and holds almost all weather, clouds, and storms.
- Thickness: about **8 km at poles** to **16–18 km at the equator**.
- Temperature generally **decreases with height** (environmental lapse rate ≈ **6.5°C/km**).
- Contains most of the atmosphere's mass and nearly all water vapour.
- Ends at the **tropopause**, where the fall in temperature stops.

- Thickness is greatest in the tropics (~**18–20 km**) and least near the poles; average height is often cited near **~13 km**.
- UP Indo-Gangetic plains sit in the dense lower troposphere — winter **inversions** trap smog (Kanpur (कानपुर), Lucknow (लखनऊ), Ghaziabad).

> **Logic:** Weather = troposphere only. Ozone shield ≠ troposphere. Homosphere excludes **ionosphere**.

35.5 Stratosphere

Identity: Ozone layer | Temp increases with height | UV shield

- The **stratosphere** lies roughly **12–50 km** above the surface.
- Temperature **increases with height** because ozone absorbs ultraviolet radiation.
- The **ozone layer (ozonosphere)** of maximum concentration sits mainly around **15–30/35 km** in the stratosphere (~**90%** of atmospheric ozone; ~**10%** in troposphere).
- Stratospheric ozone is naturally regulated in part by **nitrogen dioxide (NO_2)** and protects life from harmful **UV-B/C**; tropospheric ozone is a pollutant.
- Ends at the **stratopause**.

PYQ — UPPCS Prelims 2023, Q52

The ozone layer, which absorbs ultraviolet radiation, exists in which layer of the atmosphere?

Options: A. Troposphere B. Mesosphere C. Stratosphere D. Thermosphere

Show answer

Ans: C

PYQ — UPPCS Prelims 2018, Q52

The maximum concentration of Ozone is found in which of the following?

Options: A. Troposphere B. Mesosphere C. Stratosphere D. Exosphere

Show answer

Ans: C

PYQ — UPPCS Prelims 2018, Q57

Which pair is NOT correctly matched?

Show answer

Ozone layer — Troposphere is the wrong pair. Ozone layer = **stratosphere**.

PYQ — UPPCS Prelims 2025, Q45

With reference to the ozone layer: 1. Protects Earth's surface from solar ultraviolet radiation. 2. Ozone depletion (ओजोन क्षरण) has been linked to CFCs.

Show answer

Both correct — full ODS/Montreal depth in Topic 36.

> **Logic:** Ozone max = **stratosphere**. Pair "ozone–troposphere" is always wrong.

35.6 Mesosphere

Identity: ~50–85 km | Coldest layer | Meteors burn

- The **mesosphere** sits above the stratosphere, roughly **50–85 km**.
- Temperature **decreases with height**; the **mesopause** is the **coldest** part of the atmosphere (~**–90°C**).
- Most **meteors burn up** here due to friction with sparse air.
- Noozone-layer role and little weather activity compared with the troposphere.

> **Logic:** Meteors = mesosphere; ozone ≠ mesosphere.

35.7 Thermosphere

Identity: Very high temperature | Low density | Aurora / ISS / ionosphere

- The **thermosphere** extends roughly **85–600 km**.
- Kinetic temperature is very high, but density is so low that you would not "feel" hot.
- Contains much of the **ionosphere**, which reflects radio waves and hosts **auroras**.
- The **International Space Station** orbits in the lower thermosphere region.
- Temperature generally **increases with height** due to absorption of high-energy solar radiation.

> **Logic:** Hot ≠ dense. Thermosphere heat is not like tropospheric heat you feel.

35.8 Exosphere

Identity: Outermost layer | Merges into space | Light gases escape

- The **exosphere** is the outermost atmospheric region (~**600 km** upward, gradually fading).
- Gas molecules can travel long paths and eventually **escape** to space (especially H and He).
- It merges into interplanetary space rather than ending at a sharp boundary.
- Highest satellite orbits relate to this outer region matching.

> **Logic:** Exosphere = space transition, not the ozone layer.
